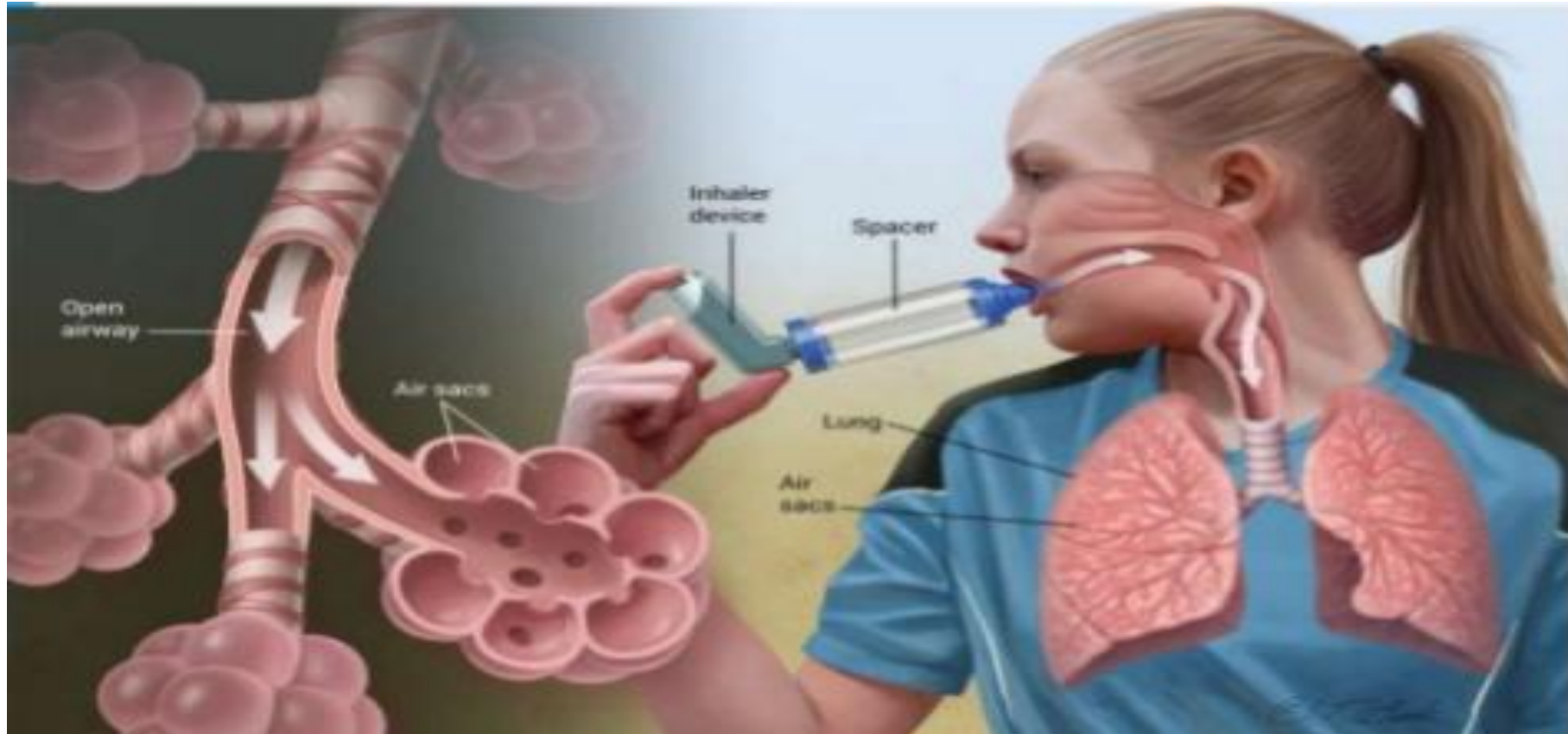


Pharmacotherapy of Asthma



By;
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Jan 2025

Learning objectives

Upon completion of the chapter, the students will be able to:

- ❖ Describe the etiology, pathophysiology and clinical presentation of acute and chronic asthma.
- ❖ List the treatment goals for asthma.
- ❖ Identify environmental factors associated with worsening asthma control.
- ❖ Select inhaled drug delivery devices based upon patient characteristics.
- ❖ Evaluate current metered-dose and dry powder inhalers technique.
- ❖ Recommend a therapeutic plan based upon asthma control and severity.
- ❖ Develop an individualized asthma action plan

Definition

- ❑GINA definition: “Asthma is a *heterogeneous* disease, usually characterized by chronic airway inflammation”.
- It is defined by the history of respiratory symptoms such as **wheeze, shortness of breath, chest tightness, and cough** that vary over time and in intensity, together with variable expiratory airflow limitation.

- Many cells and cellular elements play a role: in particular, mast cells, eosinophils, T lymphocytes, macrophages, neutrophils, and epithelial cells
- This causes obstructive airway disease which is known as bronchial asthma

EPIDEMIOLOGY

- Asthma causes for the suffering of **300 million** people and responsible for the deaths of **250,000** every year worldwide.
- Prevalence is increasing in many countries, especially in children
- Asthma is a major cause of school and work absence
- Health care expenditure on asthma is very high
 - ✓ Developed economies might expect to spend 1-2 percent of total health care expenditures on asthma.
 - ✓ Developing economies likely to face increased demand due to increasing prevalence of asthma
 - ✓ Poorly controlled asthma is expensive
 - ✓ However, investment in prevention medication is likely to yield cost savings in emergency care

Etiology

Genetic predisposition(~60-80%) plus environmental interaction play a role in the development of asthma, yet the picture remains complex and incomplete.

- Allergens
- Occupational exposures
- Viral respiratory tract infections
- Exercise
- Emotions
- Exposure to irritants
- Environmental exposures
- Drugs
 - Aspirin
 - NSAIDS
 - Cyclooxygenase-2 inhibitors (in aspirin-sensitive asthma patients)
 - Antiadrenergic and cholinergic drugs
- Preservatives
 - Tartazine, Sulfites, benzalkonium chloride



Asthma Triggers

Protective factors

Household:

- Being the younger sibling

Birth and nursing:

- Natural birth
- Breastfeeding

Farm living:

- Agriculture
- Pig/cattle farming
- Unpasteurized milk consumption
- Constant stay in animal sheds
- Silage

Microbiological exposures:

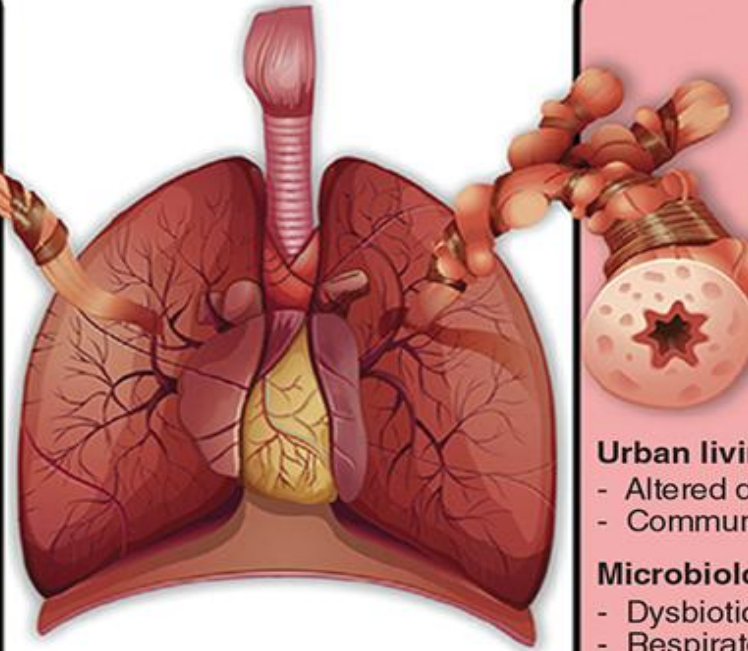
- Diverse and healthy microbiota (including members of the FLVR groups)
- Foodborne pathogens (eg, HAV, *H. pylori*)
- High-burden helminth infections (eg, *A. lumbricoides*, *T. trichiura*)

Higher socioeconomic status:

- Better access to doctors/treatments
- Increased education level
- Lower stress

Other environmental factors:

- Healthy diet
- Low pollution rates
- Exercise



ASTHMA

Risk factors

Household:

- Asthma history in the family

Birth and nursing:

- Caesarian section
- Formula feeding

Farm living:

- Sheep farming
- Pressed or loose hay

Urban living:

- Altered dietary practices
- Community associated infections

Microbiological exposures:

- Dysbiotic microbiota
- Respiratory viral infections (eg, RV, RSV)
- Bacterial pathogens (eg, *M. catarrhalis*, *S. pneumoniae*)
- Lower burden helminth infections (eg, *T. canis*)

Lower socioeconomic status:

- Increased smoking rates
- Higher stress

Other environmental factors:

- Smoking
- Obesity
- Use of antibiotics



Source: Joseph T. DiPiro, Gary C. Yee, Stuart T. Haines, Thomas D. Nolin, Vicki L. Ellingrod, L. Michael Posey: *DiPiro's Pharmacotherapy: A Pathophysiologic Approach*, 12e
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Factors that are associated with protecting against, or risk for, developing asthma. These various factors have relative degrees of importance from patient to patient. (Reprinted from van Tilburg Bernardes E, Arrieta MC. Hygiene hypothesis in asthma development: Is hygiene to blame? *Arch Med Res*. 2017;48(8):717–726.

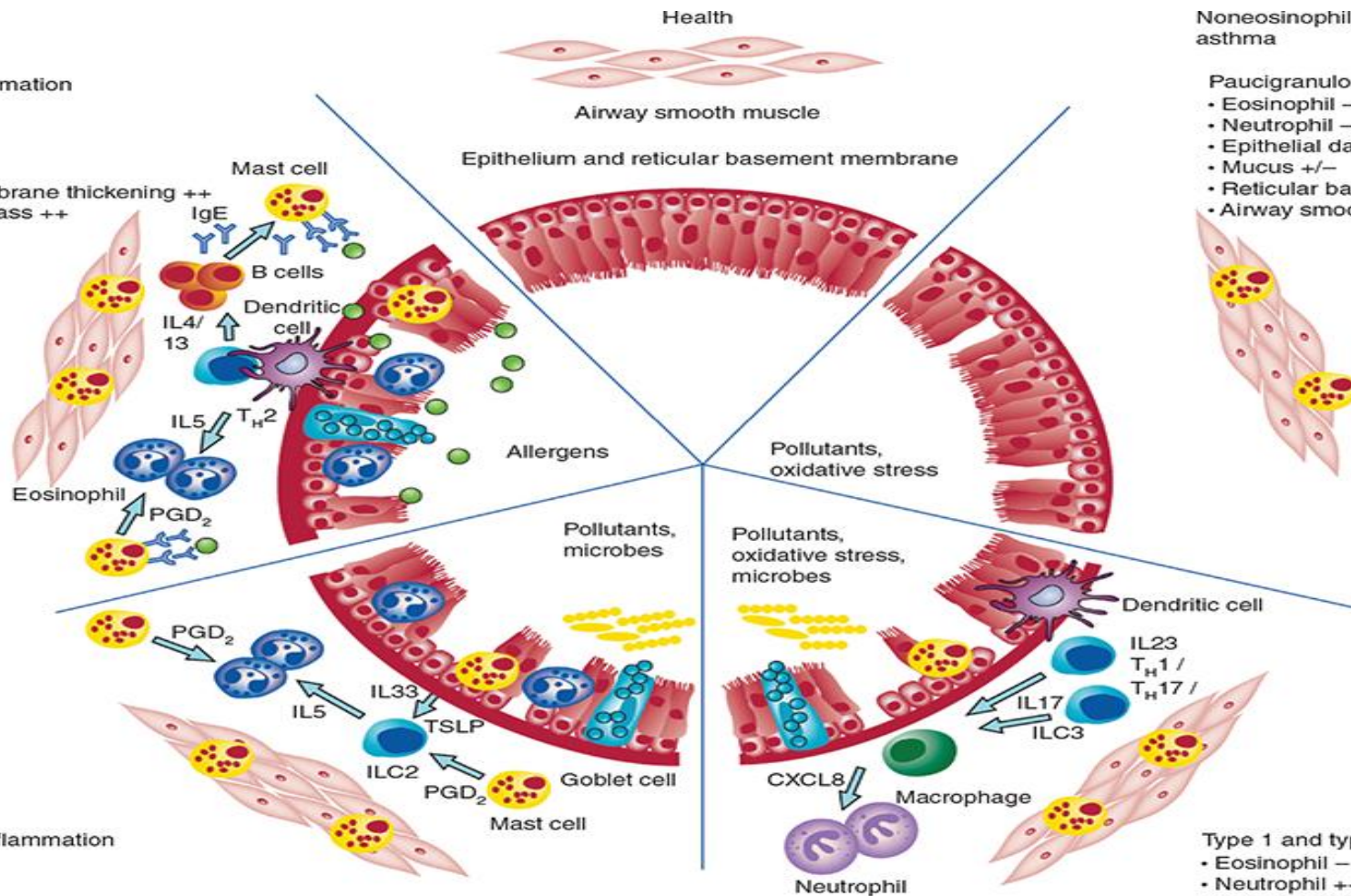
Pathophysiology

- Asthma is a complex condition where interaction of **genetics and environment** occurs involving many inflammatory cells which release a wide range of variety of mediators.
- These mediators act on the cells of the airway leading to
 - ✓ smooth muscle contraction,
 - ✓ mucus hyper secretion,
 - ✓ plasma leakage,
 - ✓ edema,
 - ✓ activation of cholinergic reflexes and activation of sensory nerves,
 - which lead to amplification of the continuing inflammatory response.

Eosinophilic asthma

Allergic eosinophilic inflammation

- Eosinophil ++
- Neutrophil –
- Epithelial damage ++
- Mucus +
- Reticular basement membrane thickening ++
- Airway smooth muscle mass ++



Nonallergic eosinophilic inflammation

- Eosinophil ++
- Neutrophil –
- Epithelial damage ++
- Mucus +
- Reticular basement membrane thickening ++
- Airway smooth muscle mass ++

Mixed granulocytic asthma

- Eosinophil ++
- Neutrophil –
- Epithelial damage ++
- Mucus ++
- Reticular basement membrane thickening +
- Airway smooth muscle +

Noneosinophilic asthma

Paucigranulocytic

- Eosinophil –
- Neutrophil –
- Epithelial damage +
- Mucus +/-
- Reticular basement membrane thickening +/-
- Airway smooth muscle mass +

Type 1 and type 17 neutrophilic inflammation

- Eosinophil –
- Neutrophil ++
- Epithelial damage ++
- Mucus ++
- Reticular basement membrane thickening +
- Airway smooth muscle mass +

Source: Joseph T. DiPiro, Gary C. Yee, Stuart T. Haines, Thomas D. Nolin, Vicki L. Ellingrod, L. Michael Posey: *DiPiro's Pharmacotherapy: A Pathophysiologic Approach*, 12e
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Diagrammatic presentation of the relationship between inflammatory cells, lipid and preformed mediators, inflammatory cytokines, and proposed pathogenesis and clinical presentation in asthma. (IL, interleukin; PG, prostaglandin; TSLP, thymic stromal lymphopoietin; CXCL8, C-X-C motif chemokine ligand 8; ILC2, type 2 innate lymphoid cells; Th, T helper.) (Reprinted from Papi A, Brightling C, Pedersen SE, Reddel HK. Asthma. Lancet. 2018;391(10122):783–800.)

Inflammatory cells

Mast cells
Eosinophils
T_H2 cells
Basophils
Neutrophils
Platelets

Structural cells

Epithelial cells
Sm muscle cells
Endothelial cells
Fibroblast
Nerves



Mediators

Histamine
Leukotrienes
Prostanoids
PAF
Kinins
Adenosine
Endothelins
Nitric oxide
Cytokines
Chemokines
Growth factors



Effects

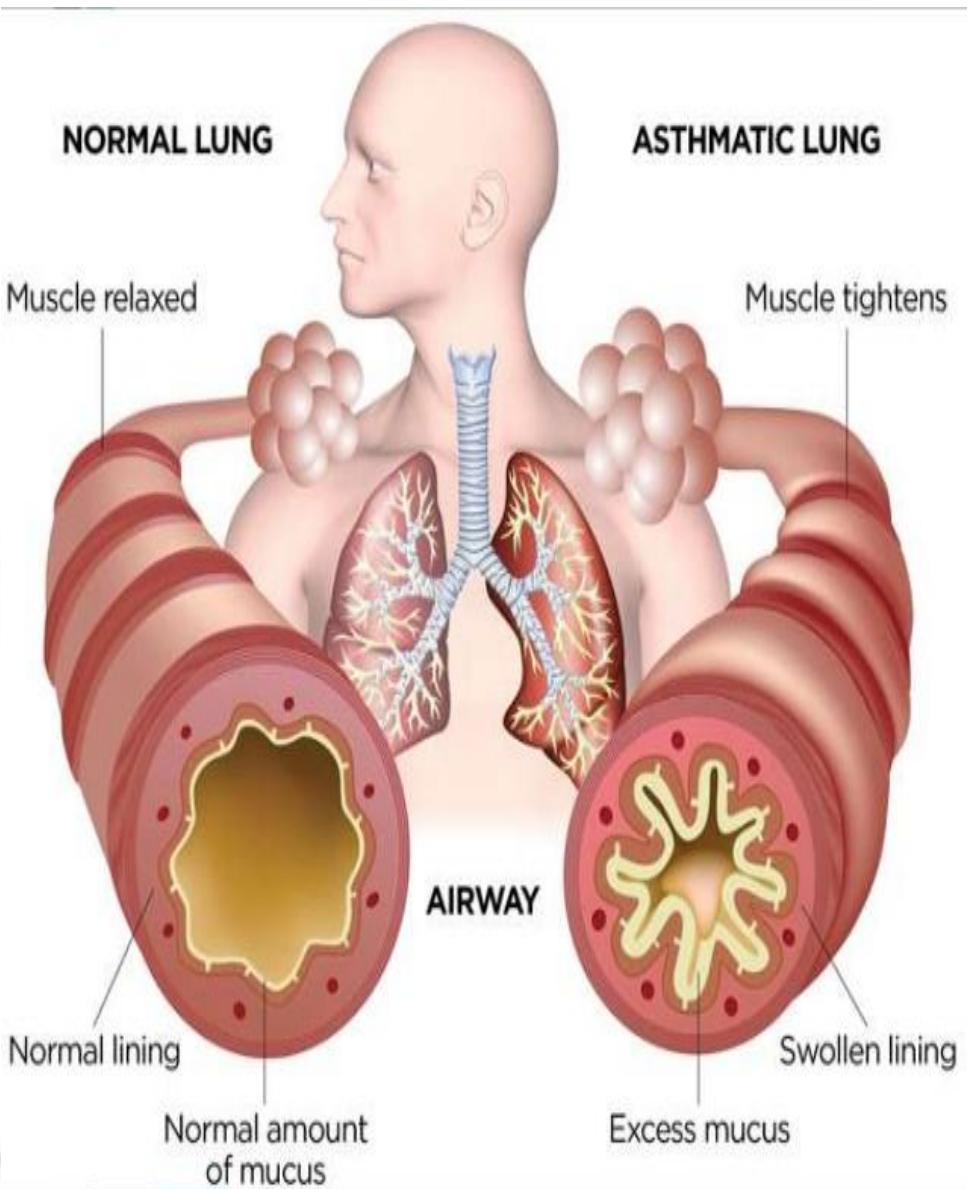
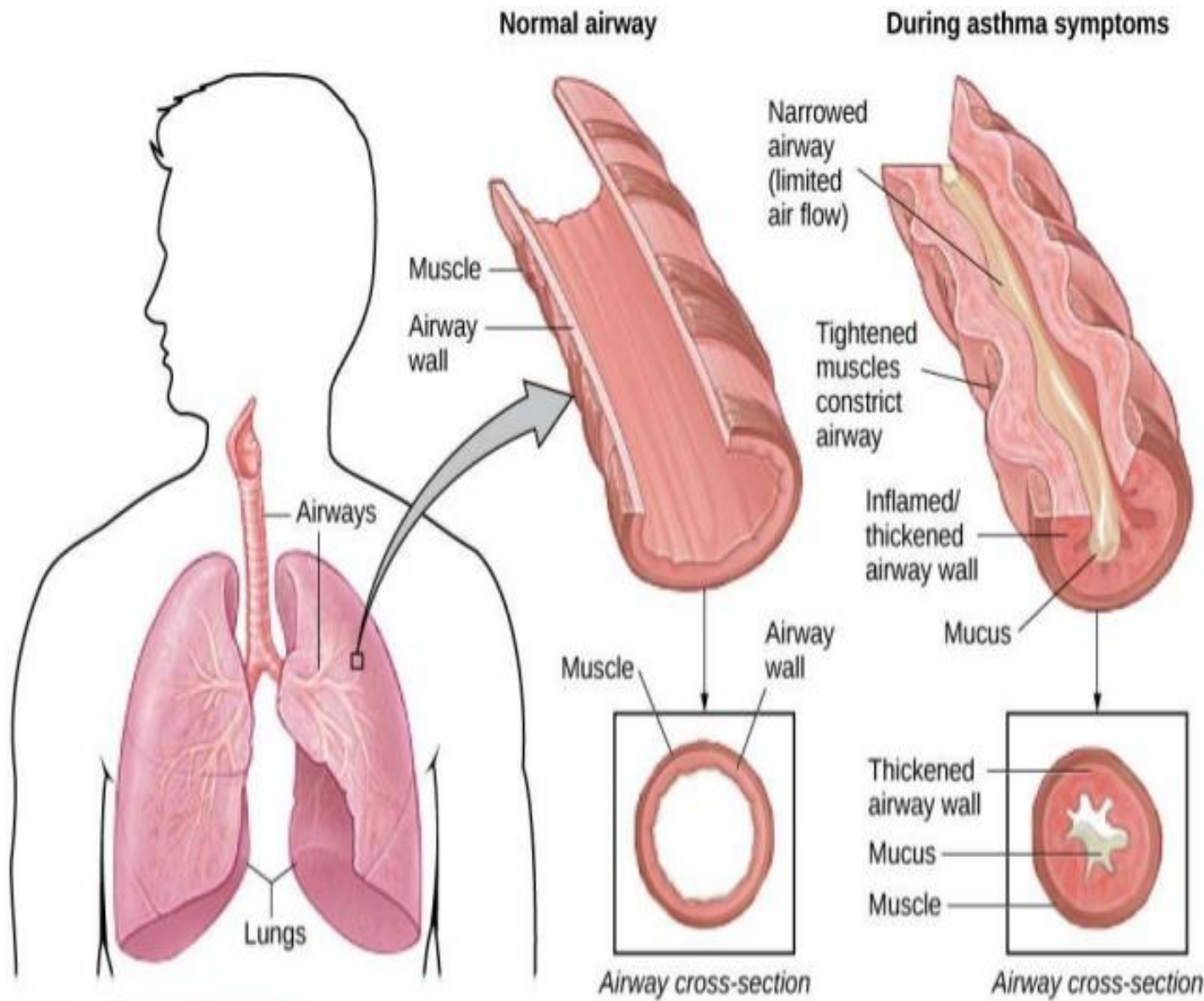
Bronchospasm
Plasma exudation
Mucus secretion
AHR
Structural changes

❑ With exposure to a trigger, a cascade of cellular responses result in:

- Increased mucus production
- Mucosal swelling
- Airway hyperreactivity

❑ Chronic inflammation also causes

- bronchoconstriction
- increased airway responsiveness
- Airway edema and remodeling and
- episodic asthma symptoms



Clinical Presentation

- Wheezing
- History of any of the following:
 - Coughing (worse particularly at nights)
 - Recurrent wheeze
 - Recurrent difficulty breathing
 - Recurrent chest tightness
- Symptoms occur or worsen at night, awakening the patient and seasonal
- Symptoms occur or worsen in the presences of:
 - Exercise
 - Viral infection
 - Inhalant allergen
 - Irritants
 - Changes in weather
 - Strong emotional expression
 - Stress
 - Menstrual cycles

Diagnosis

- ❖ Detail medical history
- ❖ Signs and symptoms
- ❖ Pulmonary function tests using Spirometry
 - Demonstrate obstruction and assess reversibility in patients ≥ 5 years
 - Forced vital capacity(FVC), forced expiratory volume in 1 second (FEV1), FEV1/FVC and peak expiratory flow rate(PEFR)
 - Ratio of the FEV1 over FVC is reviewed to determine if obstruction is present.
- ❖ Biomarkers of inflammation
- ❖ Peak flow meter
 - Used mostly for monitoring of asthma treatment effectiveness
 - Not reliable test for diagnosis because of high patients variability

Classification of Asthma Severity

Children 0-4 and 5-11 Years of Age					
		Intermittent	Persistent		
	Components		Mild	Moderate	Severe
Impairment	Symptoms	≤2 days/week	>2 days/wk but not daily	Daily	Throughout the day
	Nighttime awakenings (0-4 years)	0	1-2 × month	3-4 × month	>1 × week
	Nighttime awakenings (5-11 years)	≤2 × month	3-4 × month	>1 × wk, but not nightly	Often 7 × wk
	SABA use for symptom control	≤2 days/wk	>2 days/wk but not daily	Daily	Several times per day
	Interference with normal activity	None	Minor limitation	Some limitation	Extremely limited
	Lung function	FEV ₁ > 80%	FEV ₁ > 80%	FEV ₁ 60%-80%	FEV ₁ < 60%
	5-11 years	FEV ₁ /FVC > 85% (0.85)	FEV ₁ /FVC > 80% (0.80)	FEV ₁ /FVC 75%-80% (0.75-0.80)	FEV ₁ /FVC < 75% (0.75)
	Exacerbations	Intermittent	Persistent		
Risk	0-4 years	0-1/year	≥2 in 6 months or ≥4 wheezing episodes/1 year lasting >1 day		
	5-11 years	0-2/year	>2 in 1 year		
	Recommended initial treatment	Step 1	Step 2	Step 3 and consider short-course of oral corticosteroids	

aNormal FEV₁/FVC: 8 to 19 years 85% (0.85).

	Youths ≥12 Years of Age and Adults				
		Intermittent	Persistent		
	Components		Mild	Moderate	Severe
Impairment	Symptoms	≤2 days/week	>2 days/week but not daily	Daily	Throughout the day
	Nighttime awakenings	≤2 × month	3-4 × month	>1 × week, but not nightly	Often 7 × week
	SABA use for symptom control	≤2 days/week	>2 days/week, but not >1 × day	Daily	Several times per day
	Interference with normal activity	None	Minor limitation	Some limitation	Extremely limited
	Lung function	FEV ₁ > 80%	FEV ₁ > 80%	FEV ₁ 60%-80%	FEV ₁ < 60%
		^a FEV ₁ /FVC normal	^a FEV ₁ /FVC normal	^a FEV ₁ /FVC reduced 5% (0.05)	^a FEV ₁ /FVC reduced > 5% (0.05)
	Exacerbations	Intermittent	Persistent		
Risk		0-2/year	>2 in 1 year		
	Recommended initial treatment	Step 1	Step 2	Step 3 and consider short course of oral corticosteroids	Step 4 or 5 and consider course of oral corticosteroid

^aNormal FEV₁/FVC: 8 to 19 years 85% (0.85); 20 to 39 years 80% (0.80); 40 to 59 years 75% (0.75); 60 to 80 years 70% (0.70).

Goals of Therapy

□ Reduce impairment

- ❖ Prevent chronic and troublesome symptoms (e.g., coughing or breathlessness in the daytime, in the night, or after exertion).
- ❖ Require infrequent use (≤ 2 days a week) of inhaled SABA for quick relief of symptoms (not including prevention of exercise-induced bronchospasm [EIB]).
- ❖ Maintain (near) normal pulmonary function.
- ❖ Maintain normal activity levels (including exercise and other physical activity and attendance at school or work).
- ❖ Meet patients' and families' expectations of and satisfaction with asthma care.

□ Reduce risk

- Prevent recurrent exacerbations of asthma and minimize the need for ED visits or hospitalizations.
- Prevent loss of lung function; for children, prevent reduced lung growth.
- Provide optimal pharmacotherapy with minimal or no adverse effects of therapy.

Comprehensive Asthma management

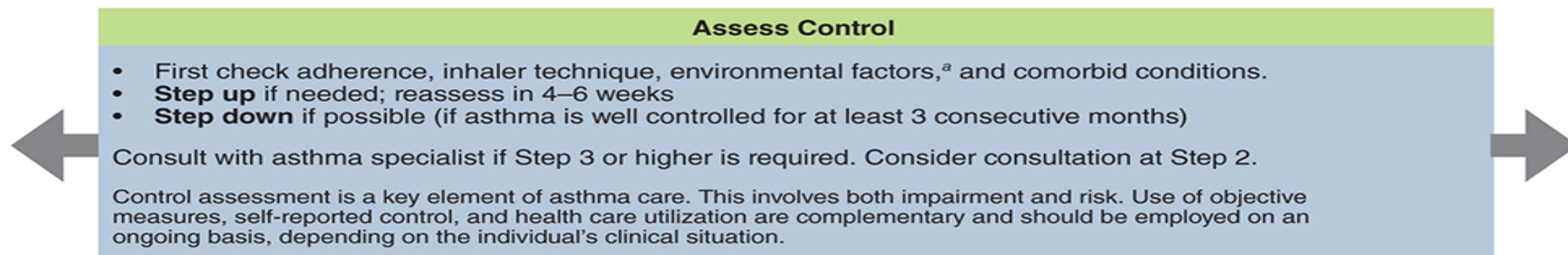
- ❖ Asthma is incurable but highly manageable condition
- ❖ 4 components of asthma management
 - ✓ Routine assessment and monitoring
 - ✓ Patient education to create a partnership between clinician and patient
 - ✓ Controlling environmental factors (trigger factors) and co-morbid conditions that contribute to asthma severity
 - ✓ Pharmacologic therapy
- ❖ Management is guided by levels of asthma severity
 - Determines regimen selection

Non-Pharmacological Management

- Humidified oxygen
- Intravenous fluids
- Environmental control
- Asthma education
- Vaccines (influenza virus, polyvalent pneumococcal)

Stepwise Approach for Managing Asthma

	Intermittent Asthma	Management of Persistent Asthma in Individuals Ages 0–4 Years				
Treatment	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6
Preferred	PRN SABA and At the start of RTI: Add short course daily ICS ^a	Daily low-dose ICS and PRN SABA	Daily medium-dose ICS and PRN SABA	Daily medium-dose ICS-LABA and PRN SABA	Daily high-dose ICS-LABA and PRN SABA	Daily high-dose ICS-LABA + oral systemic corticosteroid and PRN SABA
Alternative		Daily montelukast ^b or Cromolyn, ^b and PRN SABA		Daily medium-dose ICS + montelukast ^b and PRN SABA	Daily high-dose ICS + montelukast ^b and PRN SABA	Daily high-dose ICS + montelukast ^b + oral systemic corticosteroid and PRN SABA
			For children age 4 years only, see Step 3 and Step 4 on Management of Persistent Asthma in Individuals Ages 5–11 Years diagram.			



Source: Joseph T. DiPiro, Gary C. Yee, Stuart T. Haines, Thomas D. Nolin, Vicki L. Ellingrod, L. Michael Posey: *DiPiro's Pharmacotherapy: A Pathophysiologic Approach*, 12e
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Stepwise approach for management of asthma in ages 0 to 4 years. (ICS, inhaled corticosteroids; LABA, long-acting β_2 -agonist; SABA, inhaled short-acting β_2 -agonist; RTI, respiratory tract infection; PRN, as needed.) ^aUpdated based on the 2020 guidelines. ^bCromolyn and montelukast were not considered for this update and/or have limited availability for use in the United States. The Food and Drug Administration (FDA) issued a Boxed Warning for montelukast in March 2020. (Data from Reference 11.)

	Intermittent Asthma	Management of Persistent Asthma in Individuals Ages 5–11 Years				
Treatment	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6
Preferred	PRN SABA	Daily low-dose ICS and PRN SABA	Daily and PRN combination low-dose ICS-formoterol ^a	Daily and PRN combination medium-dose ICS-formoterol ^a	Daily high-dose ICS-LABA and PRN SABA	Daily high-dose ICS-LABA + oral systemic corticosteroid and PRN SABA
Alternative		Daily LTRA, ^b or Cromolyn, ^b or Nedocromil, ^b or Theophylline, ^b and PRN SABA	Daily medium-dose ICS and PRN SABA or Daily low-dose ICS-LABA, or daily low-dose ICS + LTRA, ^b or daily low-dose ICS + Theophylline, ^b and PRN SABA	Daily medium-dose ICS-LABA and PRN SABA or Daily medium-dose ICS + LTRA ^b or daily medium-dose ICS + Theophylline, ^b and PRN SABA	Daily high-dose ICS + LTRA ^b or daily high-dose ICS + Theophylline, ^b and PRN SABA	Daily high-dose ICS + LTRA ^b + oral systemic corticosteroid or daily high-dose ICS + Theophylline ^b + oral systemic corticosteroid, and PRN SABA
		Steps 2–4: Conditionally recommend the use of subcutaneous immunotherapy as an adjunct treatment to standard pharmacotherapy in individuals ≥ 5 years of age whose asthma is controlled at the initiation, build up, and maintenance phases of immunotherapy ^b			Consider Omalizumab ^{c,a}	

Assess Control

- First check adherence, inhaler technique, environmental factors,^a and comorbid conditions.
- **Step up** if needed; reassess in 2–6 weeks
- **Step down** if possible (if asthma is well controlled for at least 3 consecutive months)

Consult with asthma specialist if Step 4 or higher is required. Consider consultation at Step 3.

Control assessment is a key element of asthma care. This involves both impairment and risk. Use of objective measures, self-reported control, and health care utilization are complementary and should be employed on an ongoing basis, depending on the individual's clinical situation.

Source: Joseph T. DiPiro, Gary C. Yee, Stuart T. Haines, Thomas D. Nolin, Vicki L. Ellingrod, L. Michael Posey: *DiPiro's Pharmacotherapy: A Pathophysiologic Approach, 12e*
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Stepwise approach for management of asthma in ages 5 to 11 years. (ICS, inhaled corticosteroids; LABA, long-acting β_2 -agonist; LTRA, leukotriene receptor antagonist; SABA, inhaled short-acting β_2 -agonist.) ^aUpdated based on the 2020 guidelines. ^bCromolyn, Nedocromil, LTRAs including montelukast, and Theophylline were not considered for this update and/or have limited availability for use in the United States. The FDA issued a Boxed Warning for montelukast in March 2020. ^cOmalizumab is the only asthma biologic currently FDA-approved for this age range. (Data from Reference 11.)

	Intermittent Asthma	Management of Persistent Asthma in Individuals Ages 12+ Years				
Treatment	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6 ^d
Preferred	PRN SABA	Daily low-dose ICS and PRN SABA or PRN concomitant ICS and SABA ^a	Daily and PRN combination low-dose ICS-formoterol ^a	Daily and PRN combination medium-dose ICS-formoterol ^a	Daily medium-high dose ICS-LABA + LAMA and PRN SABA ^a	Daily high-dose ICS-LABA + oral systemic corticosteroids + PRN SABA
Alternative		Daily LTRA ^b and PRN SABA or Cromolyn, ^b or Nedocromil, ^b or Zileuton, ^b or Theophylline, ^b and PRN SABA	Daily medium-dose ICS and PRN SABA or Daily low-dose ICS-LABA, or daily low-dose ICS + LAMA, ^a or daily low-dose ICS + LTRA, ^b and PRN SABA or Daily low-dose ICS + Theophylline ^b or Zileuton, ^b and PRN SABA	Daily medium-dose ICS-LABA or daily medium-dose ICS + LAMA, and PRN SABA ^a or Daily medium-dose ICS + LTRA, ^b or daily medium-dose ICS + Theophylline, ^b or daily medium-dose ICS + Zileuton, ^b and PRN SABA	Daily medium-high dose ICS-LABA or daily high-dose ICS + LTRA, ^b and PRN SABA	
		Steps 2–4: Conditionally recommend the use of subcutaneous immunotherapy as an adjunct treatment to standard pharmacotherapy in individuals ≥ 5 years of age whose asthma is controlled at the initiation, build up, and maintenance phases of immunotherapy ^a			Consider adding Asthma Biologics (e.g., anti-IgE, anti-IL5, anti-IL5R, anti-IL4/IL13) ^c	

Assess Control

- First check adherence, inhaler technique, environmental factors,^a and comorbid conditions.
- **Step up** if needed; reassess in 2–6 weeks
- **Step down** if possible (if asthma is well controlled for at least 3 consecutive months)

Consult with asthma specialist if Step 4 or higher is required. Consider consultation at Step 3.

Control assessment is a key element of asthma care. This involves both impairment and risk. Use of objective measures, self-reported control, and health care utilization are complementary and should be employed on an ongoing basis, depending on the individual's clinical situation.

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Stepwise approach for management of asthma in ages 12 years and older. (ICS, inhaled corticosteroids; LABA, long-acting β_2 -agonist; LAMA, long-acting muscarinic antagonist; LTRA, leukotriene receptor antagonist; SABA, inhaled short-acting β_2 -agonist.) ^aUpdated based on the 2020 guidelines. ^bCromolyn, nedocromil, LTRAs including zileuton and montelukast, and theophylline were not considered for this update and/or have limited availability for use in the United States, and/or have an increased risk of adverse consequences and need for monitoring that makes their use less desirable. The FDA issued a Boxed Warning for montelukast in March 2020. ^cThe AHRQ systematic reviews that informed this report did not include studies that examined the role of asthma biologics (eg, anti-IgE, anti-IL5, anti-IL5R, anti-IL4/IL13). Thus this report does not contain specific recommendations for the use of biologics in asthma in Steps 5 and 6. ^dData on the use of LAMA therapy in individuals with severe persistent asthma (Step 6) were not included in the AHRQ systematic review and thus no recommendation is made. (Data from Reference 11.)

Asthma management

- Regimen selection is based on severity
- After this, the levels of control should be assessed and step down and step up therapy should be done
 - Adequately controlled
 - Step down treatment
 - Partially controlled and uncontrolled
 - Step up therapy
- The goals of therapy are to achieve and maintain adequate control

Levels of asthma control

- ❖ Should done every clinic visits and preferably for every 4 weeks helps for stepping up or dawn the treatment to achieve the control

Characteristics	Controlled(all of the following)	Partly controlled	uncontrolled
Day time symptoms	None(less 2x/wk)	More than 2x/wk	Three or more features of partly controlled asthma
Limitation of activities	none	any	
Nocturnal awakenings	none	any	
Frequency of reliever use	None/ two or less/wk	More than 2x/wk	
Lung function (PEF Or FEV1)	Normal	Less than 80% of predicted or personal best	

Asthma Medications

❖ Quick-relief asthma medications

- Systemic corticosteroids
- Short-acting β_2 agonist
- Anticholinergics

❖ Long-term asthma medications

- Oral and inhalational corticosteroids
- Inhaled long-acting β_2 agonist
- Cromolyn/Nedocromil
- Leukotriene modulators
- Methylxanthine
- Immunomodulators

Corticosteroids

- MOA: corticosteroids ***block late reaction to allergen*** and ***reduce airway hyperresponsiveness***.
 - They inhibit cytokine production, adhesion protein activation and inflammatory cell migration and activation.
- Route of administration (systemic, intravenous, oral or inhaled) is determined by the condition of the patient.
- S/S: facial flushing, appetite stimulation, GI irritation, headache, mood changes, acne exacerbation, weight gain, hyperglycemia, leukocytosis, hypokalemia
- DI: enzyme inducers (rifampin, barbiturates), estrogens, oral contraceptives, itraconazole, macrolide antibiotics, cyclosporine and potassium depleting diuretics.

Systemic Corticosteroids

- Used for rapid response during an exacerbation
- Improves pulmonary function within 1-3 hr
- Maximum effect is not achieved until 6 -9 hours or longer after administration.
- Patients should receive regular ophthalmological evaluations and osteoporosis screening and preventative therapy(Ca^{2+} , Vitamin D) if indicated.
- **Caution:** in patients who have diabetes, hypertension, adrenal suppression, congestive heart failure, PUD, osteoporosis, candidiasis and glaucoma

Systemic Corticosteroids

Generic name	Dosage form	Usual dosage
Methylprednisolone	Tablet	1 mg/kg/day
Prednisone	Tablets	1 mg/kg/day
Prednisolone	Tablets	1 mg/kg/day

Inhaled Corticosteroids

- Indicated for chronic treatment of asthma
- Local AEs: hoarseness and fungal infection of the mouth and throat.
 - Can be lessened through mouth rinsing, use of a spacer, and use of certain inhaled steroid products (Pulmicort Flexhaler® and beclomethasone (Qvar®))

Inhaled Corticosteroids

Generic Name	Brand Name	Strength	Dosage Form	Usual Dosage
Beclomethasone HFA	QVAR	40 mcg/puff, 80 mcg/puff	MDI	80 – 480 mcg/day
Budesonide	Pulmicort	200 mcg/inhalation	DPI (flexhaler)	1-3 inhalations/day
Budesonide + Formoterol	Symbicort	Budesonide: 80 or 160 mcg; Formoterol: 4.5 mcg	MDI	2 puffs twice daily
Budesonide	Respules	0.25 mg, 0.50 mg	Nebulizer	0.5-2 mg/day
Flunisolide	Aerobid	250 mcg/puff	MDI	1-8 puffs/day
Fluticasone	Flovent	44 mcg/puff, 110 mcg/puff, 220 mcg/puff	MDI	88-660 mcg/day
Fluticasone	Flovent Diskus	50 mcg/puff, 100 mcg/puff, 250 mcg/puff	DPI	100-500 mcg/day
Fluticasone + Salmeterol	Advair Diskus (100, 250,500)	Fluticasone: 100, 250 or 500 mcg Salmeterol: 50 mcg	DPI	1 inhalation twice daily
Fluticasone + Salmeterol	Advair HFA (45, 115, 230)	Fluticasone: 45, 115 or 230 mcg Salmeterol: 21 mcg	MDI	2 inhalation/dose twice daily
Mometasone	Asmanex Twisthaler	220 mcg/inhalation	DPI	1-2 inhalations/day at bedtime
Triamcinolone	Azmacort	75 mcg/puff	MDI/spacer	4-20 puffs/day

Beta₂-adrenergic agonists

- ❖ Relieve bronchoconstriction during acute asthma exacerbations as well as during chronic therapy and prevent exacerbations from occurring during exercise.
- ❖ MOA: stimulate beta₂ receptors, activating adenyl cyclase, which increase intracellular production of cAMP resulting in bronchodilation.
- ❖ AEs: tremor, palpitations, tachycardia, nervousness, headache, **hypokalemia** and tachyphylaxis
- ❖ Monitor: in elderly patients, hyperthyroidism, diabetes, seizures, arrhythmias and CAD

Beta₂-adrenergic agonists

❑ Short-acting beta₂ agonist

- Albuterol, levalbuterol, pirbuterol
- Reserved for worsening symptoms, **treatment** of acute exacerbations and **prophylaxis** of exercise-induced bronchospasm (EIB).

❑ Long-acting beta₂ agonist

- Salmeterol, formoterol
- Used for maintenance **treatment** of moderate and severe persistent asthma in combination with inhaled corticosteroids, **prophylaxis** of EIB and in patients with concurrent COPD
- Regimens must include a short-acting agent for the treatment of acute symptoms and also include a ICS unless when used to prevent EIB

Beta₂-adrenergic agonists

Generic Name	Brand Name	Dosage Form	Usual Dosage
Short-acting inhaled beta₂ agonists			
Albuterol HFA	Ventolin. Proventil, Proair	•MDI •Nebulizer solution	•2 puffs Q4h prn •2.5 mg Q4h prn
Pirbuterol	Maxair Autohaler	MDI	2 puffs Q4h prn
Long-acting inhaled beta₂ agonists			
Formoterol	Foradil Aerolizer	DPI	1 inhalation twice daily
Salmeterol	Serevent	DPI	1 inhalation twice daily

Leukotriene Modifiers

❖ Used in the prevention of allergen induced bronchoconstriction

▪ Montelukast, Zafirlukast

- MOA: leukotriene **receptor antagonists** that prevent leukotrienes from interacting with their receptors
- **AEs**: headache, dizziness, dyspepsia, nausea, diarrhea
- **DIs**: aspirin, erythromycin, theophylline, warfarin

▪ Zileuton

- MOA: blocks the effect of 5-**lipo-oxygenase** and ultimately blocks leukotriene production.
- **AEs**: headache, abdominal pain, **asthenia**, nausea, dyspepsia and **myalgia**
- **Precaution**: **hepatic impairment**, alcoholism
- **Monitor**: liver dysfunction, if ALT > 5x upper limit of normal: discontinue
- **DIs**: propranolol, theophylline, warfarin

Leukotriene Modifiers

Generic Name	Indication	Dosage Form	Usual Dosage
Montelukast (Singulair®)	Ages ≥ 12 months	•Oral granules	•4 mg (Age:12-23 months) every night
		•Chewable tablet	•4 mg (Age: 2-5 years) every night
		•Chewable tablet	•5 mg (Age:6-14 years) every night
		•Tablet	•10 mg (Adults) every night
Zafirlukast (Accolate®)	Ages > 5 years	Tablet	600 mg twice daily
Zileuton (Zyflo®, Zyflo CR®)	Ages ≥ 12 years	Tablet	Zyflo: 600mg four times daily
			Zyflo CR: 600mg twice daily

Mast cell stabilizers

- Cromolyn sodium (Intal®) and nedocromil sodium (Tilade®)
- Used to prevent the **early and late** response of asthma and as **maintenance therapy** to suppress nonspecific airway reactivity.
- MOA: act locally by stabilizing mast cells and thereby inhibiting mast cell degranulation
- Dose
 - Cromolyn sodium: MDI (0.8mg/puff): 2 puffs QID; nebulizer (20mg/ampule): 1 ampule QID
 - Nedocromil sodium: MDI (1.75mg/puff): 2 puffs QID
- **Not effective during an acute asthma exacerbation**
- AEs: wheezing, coughing, nasal congestion and irritation or dryness of the throat.

Methylxanthines

❑ Theophylline (Uniphyll®)

- ❖ Considered when beta-agonists and corticosteroids fail to control acute asthma exacerbations and as an alternative to LABA in persistent asthma
- ❖ MOA: inhibits phosphodiesterase enzyme resulting in increased levels of cAMP
- ❖ Dose: 10mg/kg/day up to 300 mg maximum
- ❖ AEs: nausea, vomiting, diarrhea, anorexia, palpitations, insomnia, seizures and reduced control of GERD
- ❖ Monitoring: PUD, hepatic disease, hypoxemia, hypertension, CHF, alcoholism and in the elderly

Theophylline: Therapeutic Drug Monitoring

- Therapeutic effect is achieved and toxicity is minimized by keeping drug concentrations at 5 – 15 mcg/mL
- Signs of toxicity: Tachycardia, headache, vomiting, seizures, arrhythmias

Anticholinergics

- ❑ Ipratropium bromide (Atrovent®), Ipratropium + albuterol (Combivent®)
- MOA: block postganglionic muscarinic receptors in the airway leading to reduced intrinsic vagal tone to the airways
- Dose:
 - Atrovent: 2 puffs of the MDI every 6 hours; 0.25 mg nebulizer solution every 6 hours
 - Combivent: 2 puffs of the MDI every 6 hours; 3 mL nebulizer solution every 6 hours
- AEs: Drying of mouth and respiratory secretions, blurred vision if sprayed in eyes.

Immunomodulators

❑ Omalizumab (Xolair®)

- Used in **moderate and severe asthma** in patients who are poorly controlled with **conventional therapy**
- MOA: attaches to free circulating IgE to prevent it from binding to mast cells thus inhibiting part of the inflammatory process.
- Dose: 150-375 mg SC every 2 – 4 weeks
- AEs: injection site reaction
- Black Box Warning(BBW): Anaphylaxis
 - Signs of anaphylaxis including bronchospasm, hypotension, syncope, urticaria, and tongue/throat angioedema

Special Populations

❖ Exercise-Induced Bronchospasm (EIB)

- Warm-up period may be helpful in preventing EIB
- Can be prevented with one of the following options:
 - SABA: at least 15 minutes before exercise may be helpful for 2 – 3 hours; drug of choice
 - LABA: 30 – 60 minutes before exercise
 - Cromolyn sodium and nedocromil: at least 10 minutes before exercise, duration 1-2 hours
 - Montelukast: $\geq$ 2 hours before exercise

Special Populations

- Pregnancy
 - Monitor the level of asthma control and lung function during prenatal visits.
 - **Albuterol** is the preferred SABA.
 - Inhaled corticosteroids are the preferred long-term control medication.
 - **Budesonide is the preferred ICS and has pregnancy category B designation**

Types of asthma devices

- **Inhalers** are small, handheld devices that deliver a puff of medicine into the airways
- Two basic types: metered-dose inhalers (MDIs) and dry powder inhalers (DPIs).
 - **MDIs** contain a **liquid medication** delivered as an aerosol spray. You can place the mouthpiece and breathe in slowly as you press down on the inhaler.
 - use a spacer (a hollow plastic tube between the mouthpiece and the canister of medicine)
 - The medication is release by propellant
 - Shaking is required before use

These are the general steps for using a metered dose inhaler

1. Remove the cap from the metered dose inhaler.
2. Shake the inhaler for a few seconds.
3. Place your index finger on top of the canister and thumb on the bottom of the mouthpiece.
4. Tilt your head back slightly and breathe out.
5. Hold the inhaler upright about the width of two fingers from your mouth.
6. Breathe in and out slowly through your mouth one time.
7. Press down on the inhaler as you breathe in as slowly and deeply as you can - about 3 to 5 seconds.
8. If possible, hold your breath for at least 10 seconds.
9. If more than one puff is required, wait about 1 minute and repeat steps 2-8.
10. Replace the cap on the metered dose inhaler.
11. Gargle and rinse your mouth with water or mouthwash (usually advised only for steroid-type inhalers)



.....cont

A **DPI** is similar, but it releases a puff of dry powder instead of a liquid mist.

- a spacer is not required
- Instead, close your mouth tightly around the mouthpiece of the DPI inhaler and inhale rapidly
- Never shake and breathing to the medication will remove some of the medication
- Deep breath is required to release the medication

Nebulizers are machines that convert a liquid medicine into a mist that you inhale into your lungs.

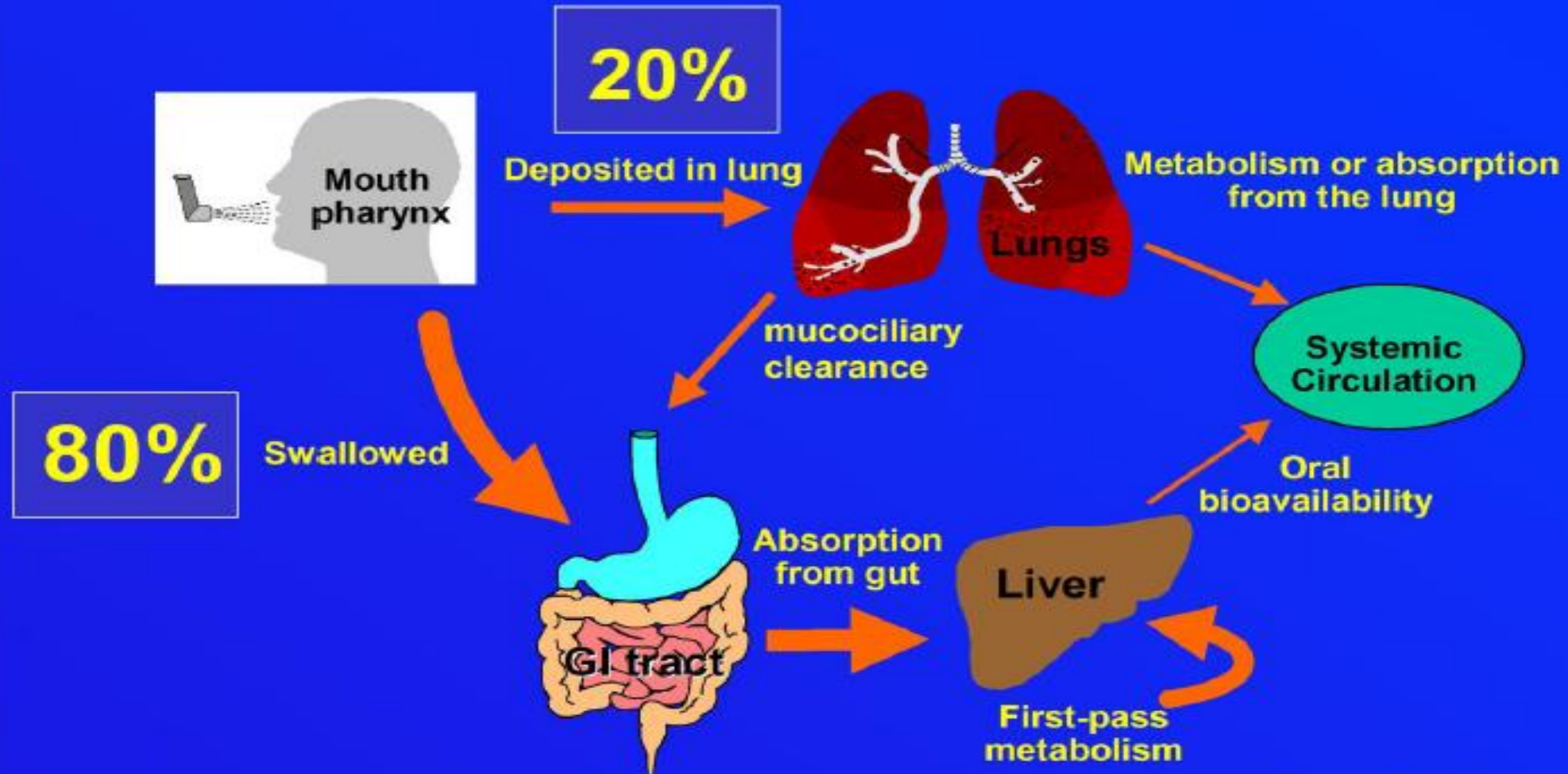
- Not portable

Suggested steps for using Diskus DPI

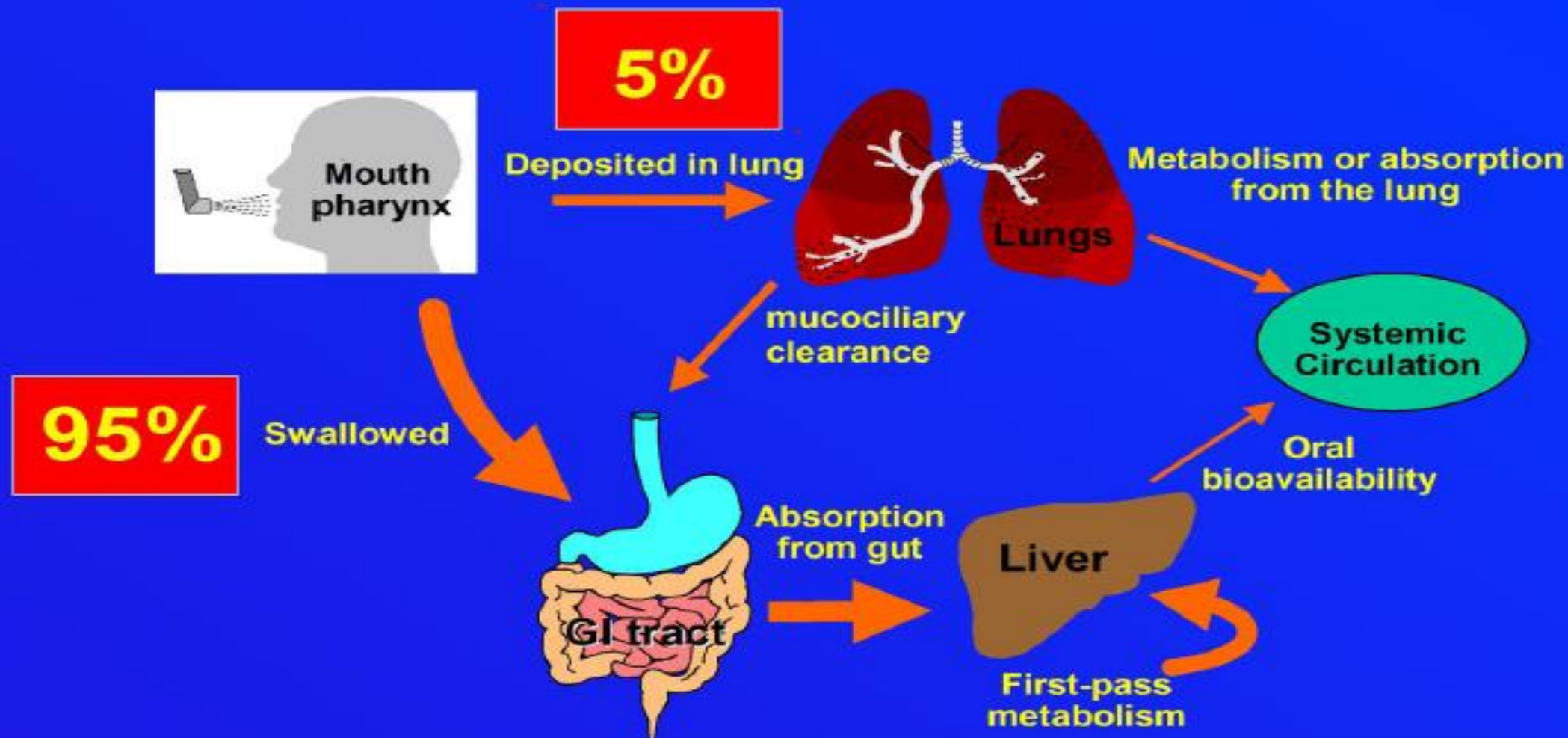
1. Check dose counter
2. Open using thumb grip
3. Holding horizontally, load dose by sliding lever until it clicks
4. Breathe out gently away from mouthpiece
5. Place mouthpiece in mouth and seal lips
6. Breathe in steadily and deeply
7. Hold breath for about 10 seconds or as long as comfortable
8. While holding breath, remove inhaler from mouth
9. Breathe out gently away from mouthpiece
10. If an extra dose is needed, repeat steps 3 to 9
11. Close cover to click shut



Fate of inhaled drugs – Good Technique



Fate of inhaled drugs – **Poor** Technique



THANKS